

Proposing Real-World Application Scenarios for Automated Bug Detection in Speedruns

Bridging gameplay data, LLM analysis, and user value.

Yuto Uenaka, Yuta Ishimoto, Masanari Kondo, Yasutaka Kamei
Kyushu University

Meeting Purpose

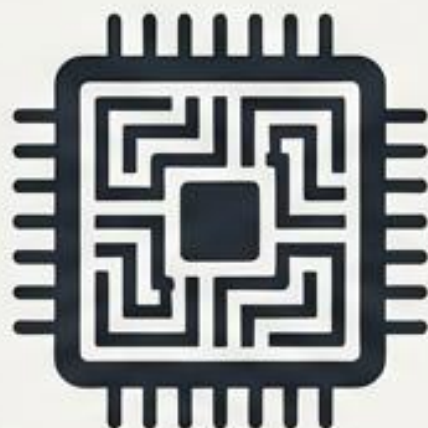
- Propose two real-world application scenarios for automated bug detection in speedruns.
- Receive feedback and solidify our research direction.

Information Architecture: The Core Engine Pipeline



Input (Raw Data)

Speedrun Gameplay Video



Processing (The Engine)

LLM-Based Analysis



Output (Structured Insight)

Extracted Bug Classifications &
Precise Timestamps
[start_time] - [end_time]

Scenario 1: Automated Organization of Optimal Speedrun Route Evolution



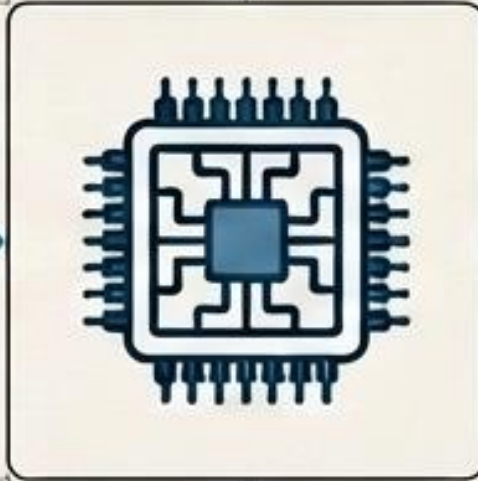
Data Source

Speedrun.com Archives



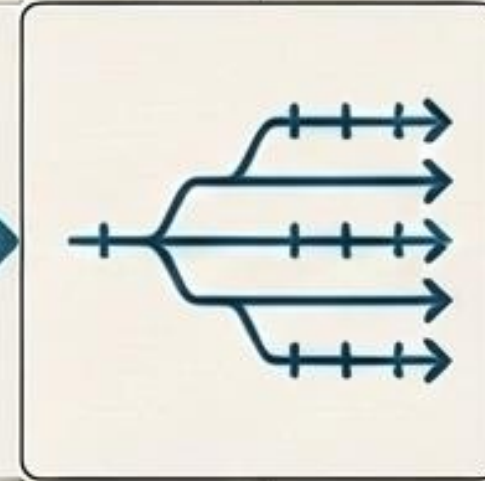
Target Selection

e.g. Super Mario 64: 120
Star - Historical World
Records



Automated Processing

Analyzing gameplay videos
to map exact timestamps to
specific bug usage across
decades of runs.



Output Generation

Chronological Route
Evolution Tree



End-User Value

Game Museum Curators /
Historians. Allows for the
automated, precise curation
of speedrun history.

Scenario 2: Automated Generation of Bug Highlight Clips

